

Nautobot Docker Deployment & Management Guide

1. System Prerequisites (Mandatory)

Before running any Docker commands, verify your environment:

- **Linux / macOS:** Docker runs natively. Ensure the Docker service is active.
 - **Windows (WSL 2 Requirement):** * Docker Desktop for Windows must use the **WSL 2** engine.
 - **Cross-Check Configuration:** Run `wsl -l -v` in PowerShell to ensure your distro (e.g., Ubuntu-20.04) is on **Version 2**.
 - **Enable Integration:** In Docker Desktop Settings, navigate to **Resources > WSL Integration** and toggle your distro to **ON**.
-

2. Fresh Installation (Scenario A)

Use this if you are starting from scratch and do not have an existing database backup.

- **Step 1: Environment Setup** Ensure a `.env` file exists with the necessary variables (e.g., `NAUTOBOT_VERSION`, `PYTHON_VER`).
 - **Step 2: Launch Containers** Run the following command to pull images and start services.

```
docker compose -f docker-compose.base.yml -f
docker-compose.postgres.yml -f docker-compose.local.yml up
-d
```
 - **Step 3: Create Admin Account** Since the database is empty, you must create a superuser to log in

```
docker exec -it environments-nautobot-1 nautobot-server
createsuperuser
```
 - Follow the prompts to set a username and password.
-

3. Migration & Data Restoration (Scenario B)

Use this if you have an existing backup (like **your Data Center**) and want to restore it into a new Docker environment.

- **Step 1: Deploy Basic Infrastructure** Start the containers using the standard command to initialize the network and volumes.

```
docker compose -f docker-compose.base.yml -f
docker-compose.postgres.yml up -d
```

- **Step 1.5: Drop & Re-create Colliding Database**

```
docker exec -it environments-db-1 psql -U autobot -d
postgres -c "DROP DATABASE autobot;"
```

```
docker exec -it environments-db-1 psql -U autobot -d
postgres -c "CREATE DATABASE autobot;"
```

- **Step 2: Transfer Backup File** (Copy your local or remote `.sql` file into the running database container.)

```
docker cp your_backup.sql environments-db-1:/tmp/backup.sql
```

- **Step 3: Ingest Data** (Overwrite the fresh database with your backup data.)

```
docker exec -t environments-db-1 psql -U autobot -d
autobot -f /tmp/backup.sql
```

- **Step 3.5: Synchronize application layer with database schema**

```
docker exec -it environments-autobot-1 autobot-server
migrate
```

```
docker exec -it environments-autobot-1 autobot-server
post_upgrade
```

- **Step 4: Sync & Restart** (Restart the application to reflect the restored data.)

```
docker restart environments-autobot-1
```

OR

- **Re-orchestrate & Apply Port Configuration**

```
docker compose -f docker-compose.base.yml -f
docker-compose.postgres.yml -f docker-compose.local.yml
down
```

```
docker compose -f docker-compose.base.yml -f
docker-compose.postgres.yml -f docker-compose.local.yml up
-d
```

4. Documentation

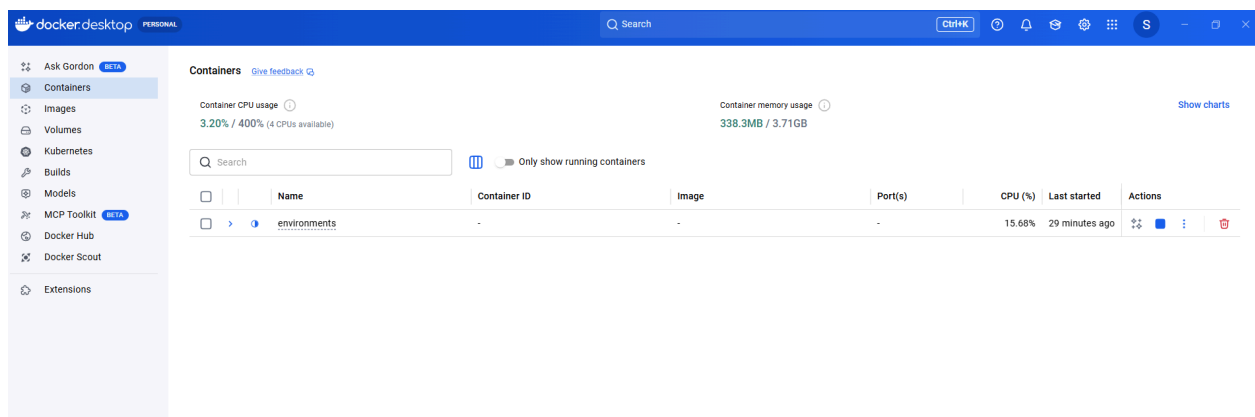
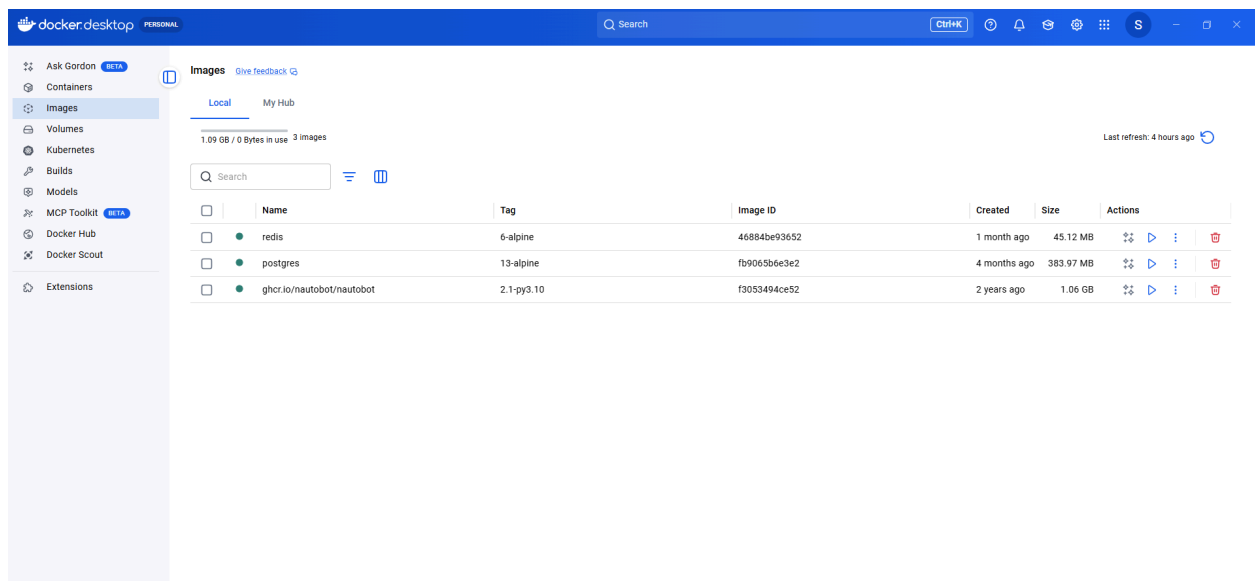
- **Build vs Pull:** If the deployment fails with a "poetry" error, it means Docker is trying to **build** from source code. Edit the `docker-compose.base.yml` to use a pre-built **image** instead.

- **Status Check:** Always use `docker ps` to ensure the database is "healthy" before attempting a restore.

Start the containers

```
docker compose -f docker-compose.base.yml -f
docker-compose.postgres.yml -f docker-compose.local.yml up -d
```

```
root@DESKTOP-6HKF3CQ:/home/huda/nautobot-docker-compose/environments#
root@DESKTOP-6HKF3CQ:/home/huda/nautobot-docker-compose/environments#
root@DESKTOP-6HKF3CQ:/home/huda/nautobot-docker-compose/environments# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                                NAMES
d8ed348e515b   ghcr.io/nautobot/nautobot:2.1-py3.10 "/docker-entrypoint.s..." 25 minutes ago Up 25 minutes (healthy) 8080/tcp, 8443/tcp environments-nautobot-1
e24fd0737e69   redis:6-alpine                      "docker-entrypoint.s..." 25 minutes ago Up 25 minutes        6379/tcp environments-redis-1
f9e119efcf7    postgres:13-alpine                 "docker-entrypoint.s..." 25 minutes ago Up 25 minutes (healthy) 5432/tcp environments-db-1
root@DESKTOP-6HKF3CQ:/home/huda/nautobot-docker-compose/environments#
root@DESKTOP-6HKF3CQ:/home/huda/nautobot-docker-compose/environments#
root@DESKTOP-6HKF3CQ:/home/huda/nautobot-docker-compose/environments#
```



dockerdesktop

PERSONAL

Q Search

Ctrl+K

S

Ask Gordon

BETA

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP Toolkit

BETA

Docker Hub

Docker Scout

Extensions

Containers

Give feedback

Container CPU usage

18.99% / 400% (4 CPUs available)

Container memory usage

1.12GB / 3.7GB

Show charts

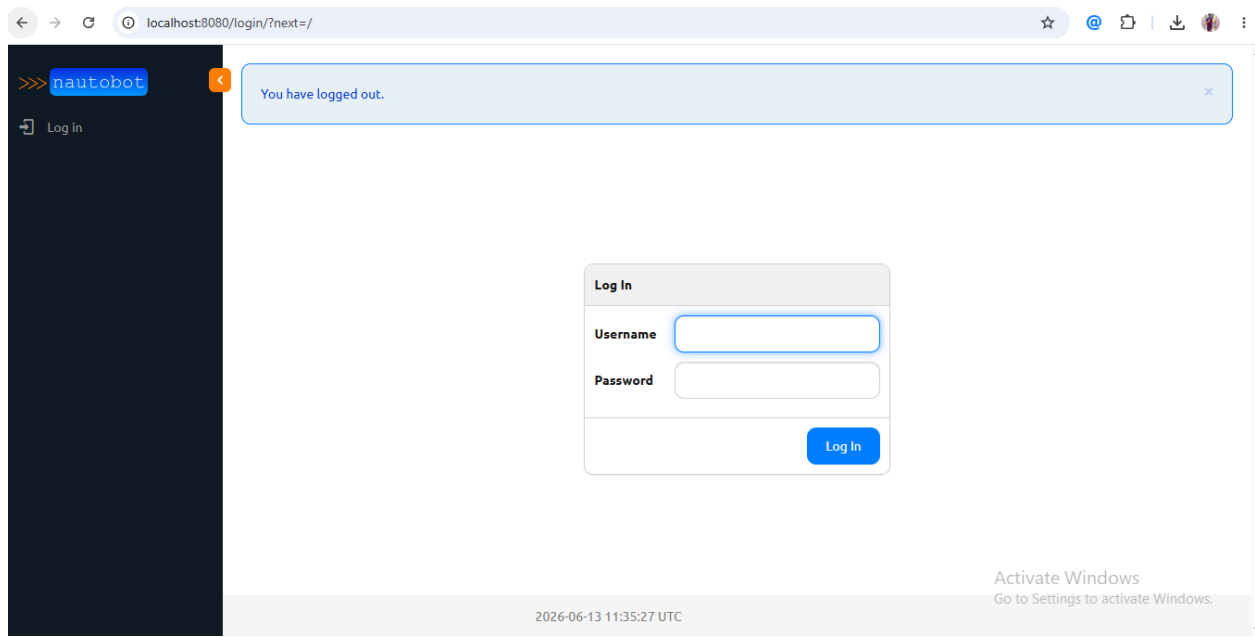
Q Search

Only show running containers

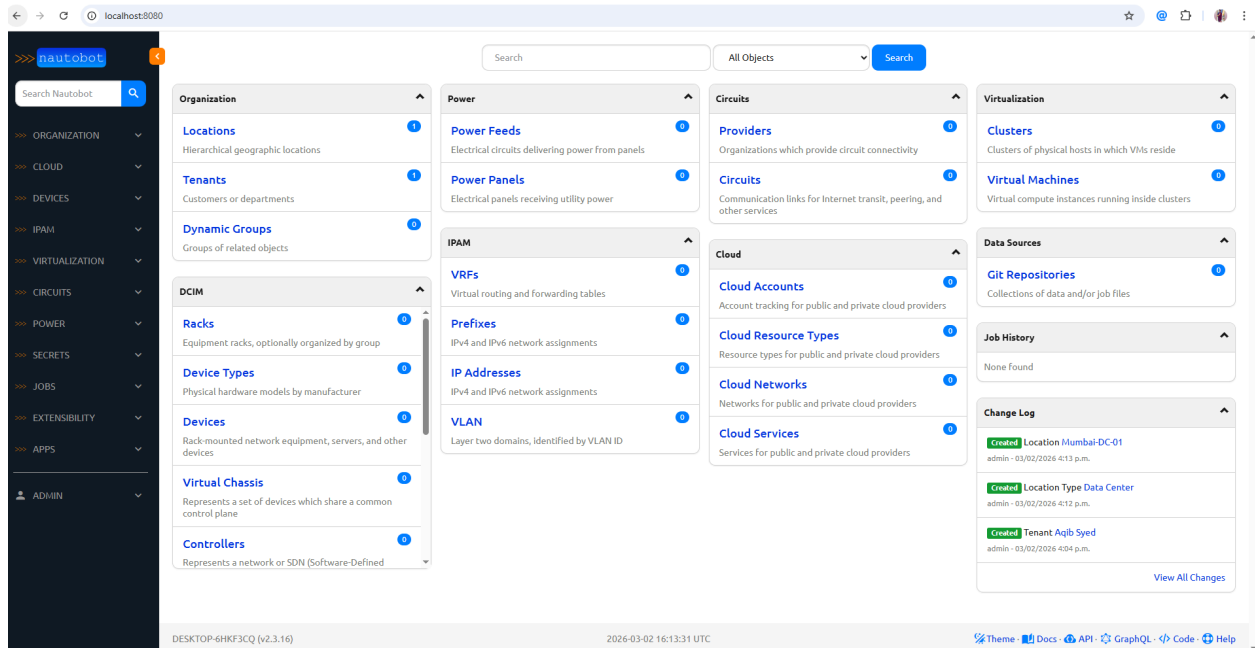
☐		Name	Container ID	Image	Port(s)	CPU (%)	Memory usag...	Actions
☐	▼	environments	-	-	-	18.92%	1.12GB / 18.97Gi	
☐		celery_worker	f1064f02e9e7	nauto		0.48%	544.1MB / 3.79G	
☐		celery_beat-1	c8238a6913a3	nauto		0%	164.1MB / 3.79G	
☐		nauto	83519e8db3d9	nauto	8080:8080	18.12%	357.4MB / 3.79G	
☐		db-1	2f75538b82f7	postgres:1		0%	76.47MB / 3.79G	
☐		redis-1	24c14b06abc8	redis:6-alpi		0.32%	5.07MB / 3.79GB	

Nautobot Production Ingestion Guide

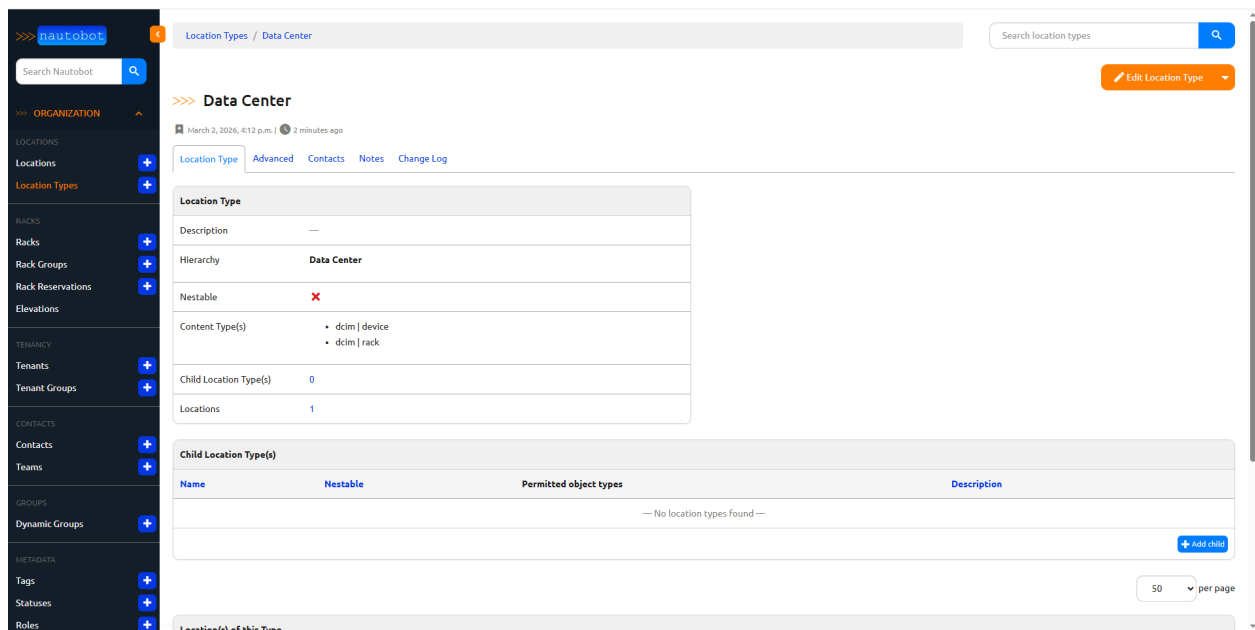
Login



Dashboard



Organization > Location Types



ORGANIZATION > Locations

Search Nautobot

ORGANIZATION

CLOUD

DEVICES

IPAM

VIRTUALIZATION

CIRCUITS

POWER

SECRETS

JOB

EXTENSIBILITY

APPS

ADMIN

Locations / Mumbai-DC-01

Search locations

Edit Location

Mumbai-DC-01

March 2, 2026, 4:13 p.m. 1 minute ago

LocationAdvancedContactsNotesChange Log

Location

Location TypeData Center

StatusActive

HierarchyMumbai-DC-01 (Data Center)

Tenant

Facility

AS Number

Time Zone

Description

Children

Geographical Info

Physical Address

Shipping Address

GPS Coordinates

Stats

0

Racks

0

Devices

0

Prefixes

0

VLANs

0

Circuits

0

Virtual Machines

Rack Groups

All racks0

Images

None

Attach an image

Tenant

Search Nautobot

ORGANIZATION

CLOUD

DEVICES

IPAM

VIRTUALIZATION

CIRCUITS

POWER

SECRETS

JOB

EXTENSIBILITY

APPS

ADMIN

Tenants / Aqib Syed

Search tenants

Edit Tenant

Aqib Syed

March 2, 2026, 4:04 p.m. 11 minutes ago

TenantAdvancedContactsNotesChange Log

Tenant

Tenant Group

DescriptionTest

Comments

None

Tags

No tags assigned

Stats

0

Circuits

0

Clusters

0

Devices

0

Dynamic groups

0

IP addresses

0

Locations

0

Prefixes

0

Racks

0

Rack reservations

0

Virtual machines

0

VLANs

0

VRFs

Step 2 - Infrastructure Setup

DEVICES > Manufacturers

>>> nautobot

Search Nautobot

ORGANIZATION

CLOUD

DEVICES

DEVICES

Devices

Virtual Chassis

Device Redundancy Groups

Interface Redundancy Groups

DEVICE TYPES

Device Types

Device Families

Manufacturers

MODULES

Modules

Module Types

SOFTWARE

Platforms

Software Versions

Software Image Files

Manufacturers / Dell

Search manufacturers

Edit Manufacturer

>>> Dell

March 2, 2026, 4:17 p.m. 9 minutes ago

Manufacturer Advanced Contacts Notes Change Log

Manufacturer

Description

Devices

Devices

Name

Status

Tenant

Role

Type

Location

Rack

IP Address

No devices found

Add device

50 per page

DESKTOP-6HKF3CQ (v2.3.16) 2026-03-02 16:27:06 UTC Theme Docs API GraphQL Code Help

DEVICES > Device Types

>>> nautobot

Search Nautobot

ORGANIZATION

CLOUD

DEVICES

DEVICES

Devices

Virtual Chassis

Device Redundancy Groups

Interface Redundancy Groups

DEVICE TYPES

Device Types

Device Families

Manufacturers

MODULES

Modules

Module Types

SOFTWARE

Platforms

Software Versions

Software Image Files

Device Types / Dell / Dell PowerEdge R740

Search device types

Add Components Edit Device Type

>>> Dell PowerEdge R740

March 2, 2026, 4:18 p.m. 10 minutes ago

Device Type Advanced Contacts Interfaces Front Ports Rear Ports Console Ports Console Server Ports Power Ports Power Outlets Device Bays Module Bays Notes Change Log

Chassis

Manufacturer

Device Family

Model Name

Part Number

Height (U)

Full Depth

Parent/Child

Front Image

Rear Image

Device Instances

Comments

None

Tags

No tags assigned

ROLES

The screenshot shows the Nautobot interface with the 'Roles' page selected. A green notification banner at the top states 'Created role Web-Server'. The left sidebar contains a navigation menu with categories like Rack Reservations, Tenancy, Contacts, Groups, Metadata, and Cloud/Devices/JPAM/Virtualization/Circuits/Power/Secrets/JOBS. The main content area displays the 'Web-Server' role configuration. It includes a breadcrumb 'Roles / Web-Server', a search bar, and an 'Edit Role' button. The role details are shown in a table:

Role	
Name	Web-Server
Weight	—
Content Type(s)	dcim device
Color	
Description	—

Below the role details is a 'Devices' table with columns: Name, Status, Tenant, Type, Location, Rack, and IP Address. The table is currently empty, showing 'No devices found'. At the bottom right, there is a '+ Add Device' button and a pagination control set to '50 per page'.

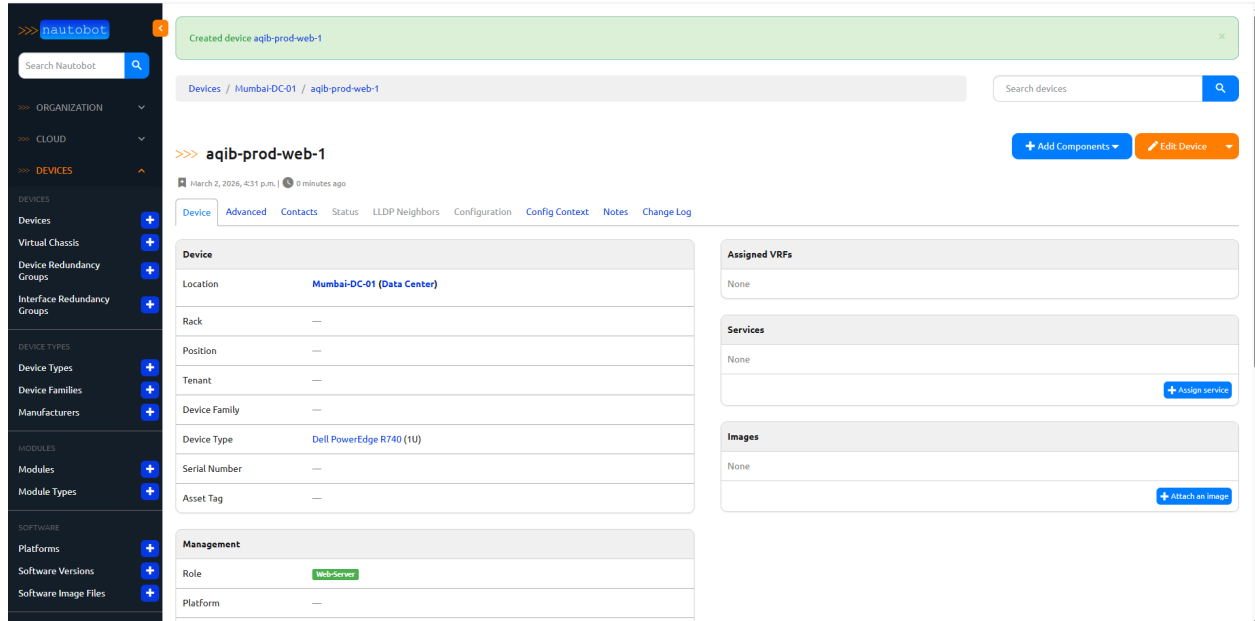
Step 3 - Add Devices and Interfaces

DEVICES > Click +ADD

The screenshot shows the Nautobot interface with the 'Devices' page selected. The left sidebar is the same as the previous screenshot. The main content area displays the 'Devices' page with a breadcrumb 'Devices', a search bar, and buttons for 'Configure', 'Filter', '+ Add', 'Import', and 'Export'. Below these is a table of devices:

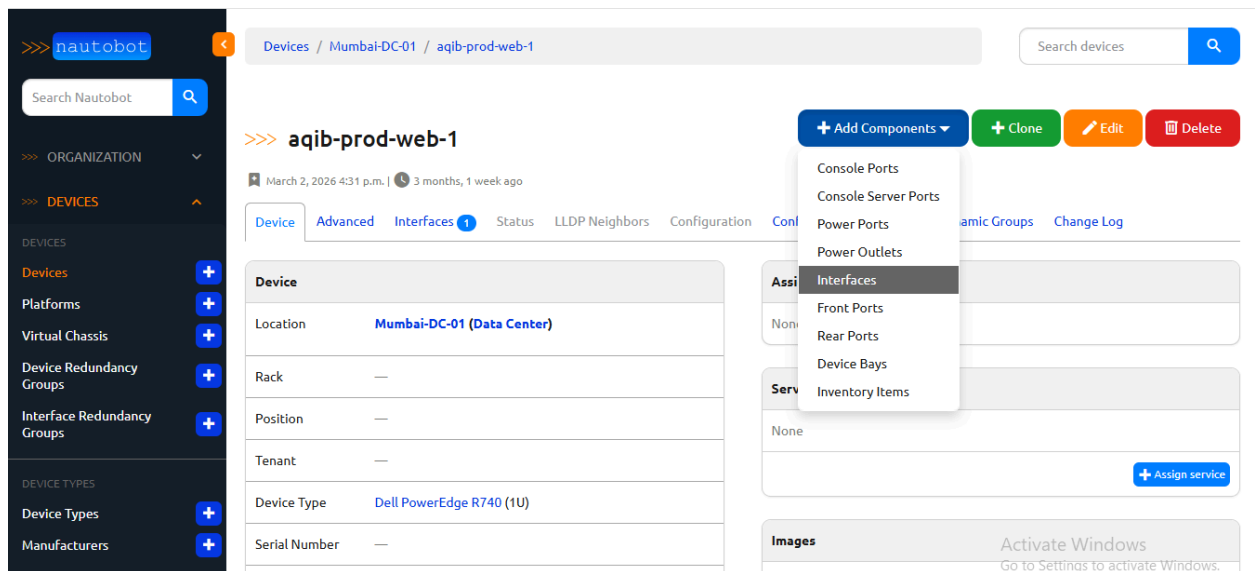
☐	Name	Status	Tenant	Role	Type	Location	Rack	IP Address
☐	aqib-prod-web-1	Active	—	Web-Server	Dell PowerEdge R740	Mumbai-DC-01	—	—

At the bottom left of the table, there are buttons: '+ Add Components', 'Edit Selected', and 'Delete Selected'. At the bottom right, there is a pagination control set to '50 per page' and the text 'Showing 1-1 of 1'.



Add Interface

Device > Select your Device > Add Components > Interfaces > Add Interface



Interface Added

>>> aqib-prod-web-1

+ Add Components

+ Clone

Edit

Delete

March 2, 2026 4:31 p.m. | 3 months, 1 week ago

DeviceAdvancedInterfaces1StatusLLDP NeighborsConfigurationConfig ContextNotesDynamic GroupsChange Log

Interfaces

Filter

Configure

	Name	Status	Label	Enabled	Type	Parent	LAG	MTU	VRF	Mode	Description	Cable	Connection	IP Addresses
☐	etho0	Active	—	✓	Virtual	—	—	—	—	—	—	—	—	+ Edit Delete

Rename

Edit

Disconnect

Delete

+ Add interfaces

>>> DEVICES

DEVICES

Devices

Virtual Chassis

Device Redundancy Groups

Interface Redundancy Groups

DEVICE TYPES

Device Types

Device Families

Manufacturers

MODULES

Modules

Module Types

SOFTWARE

Platforms

Software Versions

Software Image Files

CONTROLLERS

Controllers

Managed Device Groups

CONNECTIONS

Cables

Console Connections

Power Connections

Devices / aqib-prod-web-1 / Interfaces / etho0

>>> aqib-prod-web-1 / etho0

March 2, 2026, 4:52 p.m. | 0 minutes ago

InterfaceAdvancedContactsNotesChange Log

Interface

Device	aqib-prod-web-1
Name	etho0
Status	Active
Role	—
Label	—
Type	Virtual
Enabled	✓
Parent Interface	—
Bridge	—
LAG	—
Description	—
VRF	—
MTU	—
MAC Address	—

Click on Interface

Devices / Mumbai-DC-01 / aqib-prod-web-1

Search devices

>>> aqib-prod-web-1

+ Add Components Edit Device

March 2, 2026, 4:31 p.m.1 21 minutes ago

Device Advanced Contacts Interfaces 1 Status LLDP Neighbors Configuration Config Context Notes Change Log

Interfaces

Filter

Configure

☐	Name	Status	Role	Label	Module	Enabled	Type	Parent	LAG	MTU	VRF	Mode	Description	Cable	Connection	IP Addresses
☐	etho0	Active	—	—	—	✓	Virtual	—	—	—	—	—	—	—	—	+ Edit Delete

Rename Edit Disconnect Delete

+ Add interfaces

Assign IP

Create Prefix under IPAM

IPAM

IP ADDRESSES

IP Addresses

IP Address Assignments

PREFIXES

Prefixes

RIRs

RIRs

VRFs

Namespaces

VRFs

Route Targets

VLANs

VLANs

VLAN Groups

SERVICES

Services

VIRTUALIZATION

CIRCUITS

POWER

SECRETS

JOBS

EXTENSIBILITY

Created prefix 192.168.1.0/24

Namespaces / Global / Prefixes / 192.168.1.0/24: Global

Search prefixes

Edit Prefix

>>> 192.168.1.0/24: Global

March 2, 2026, 5:08 p.m.1 0 minutes ago

Prefix Advanced Contacts Child Prefixes 1 IP Addresses 1 Notes Change Log

Prefix

NamespaceGlobal

IP VersionIPv4

StatusActive

Role—

TypeNetwork

Tenant—

LocationsNone

VLAN—

RIR—

Date Allocated—

Description—

Utilization0%

Parent Prefixes

None

Assigned VRFs

None

Assigned Cloud Networks

None

IPAM > IP Addresses

CLOUD

DEVICES

IPAM

IP ADDRESSES

IP Addresses

IP Address Assignments

PREFIXES

Prefixes

RIRs

RIRs

VRFs

Namespaces

VRFs

Route Targets

VLANs

VLANs

VLAN Groups

SERVICES

Services

VIRTUALIZATION

CIRCUITS

POWER

IP Addresses / Global / 192.168.1.10/24

Search IP addresses

Edit IP Address

>>> 192.168.1.10/24

March 2, 2026, 5:09 p.m. 0 minutes ago

IP Address Advanced Contacts Notes Change Log

IP Address

NamespaceGlobal

IP VersionIPv4

Typehost

StatusActive

Role

VRFs

DNS Name

Description

Operational Details

Tenant

Assignment

NAT (inside)

NAT (outside)

—

—

None

None

Tags

No tags assigned

Parent Prefixes

Prefix	Type	Status	VRFs	Namespace	Tenant	Locations	VLAN	Role	Description
192.168.1.0/24	Network	Active	—	Global	—	—	—	—	—